

Supplementary Material: Majorana zero modes in silicon

W. T. Trevena

June 2026 — accompanies draft v2

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1 Model, basis, and the physical pairing channel

1.1 Single-channel BdG conventions

The normal-state Hamiltonian on N sites with spacing dx is

$$h = \sum_n c_n^\dagger [(2t - \mu + V_n)\sigma_0 + E_Z \sigma_x] c_n + \sum_n \left[c_{n+1}^\dagger (-t\sigma_0 - i a_{so} \sigma_z) c_n + \text{h.c.} \right], \quad (1)$$

with $t = \hbar^2/2m^* dx^2$, $a_{so} = \alpha/2dx$, $E_Z = \frac{1}{2}g\mu_B B$, and singlet pairing $\Delta i\sigma_y$. The BdG matrix is assembled as $H = \begin{pmatrix} h & D \\ D^\dagger & -h^* \end{pmatrix}$, $D = \Delta \mathbb{K}_N \otimes i\sigma_y$, which enforces particle-hole symmetry $\mathcal{P} = \tau_x K$ exactly on the lattice. The topological criterion at μ measured from the band bottom is $E_Z^2 > \Delta^2 + \mu^2$. All energies are handled in SI units (J) internally and quoted in μeV .

1.2 Two-valley basis and why pairing is inter-valley

The conduction-band wire has two relevant low-energy valleys at $\pm k_0 \hat{z}$, $k_0 = 0.85 (2\pi/a_{\text{Si}})$. In the envelope basis (c_+, c_-) (valley Pauli matrices ν_i), time reversal maps $+k_0 \uparrow \rightarrow -k_0 \downarrow$: *the two valleys are time-reversed partners*. A uniform s -wave parent induces pairing between time-reversed states at zero total momentum; the only zero-momentum singlet channel is therefore *inter-valley*,

$$D = \Delta (\nu_x \otimes i\sigma_y), \quad D^T = -D. \quad (2)$$

Intra-valley pairing $(c_+ c_+)$ would carry total momentum $2k_0$ (an FFLO-type condensate) and is not induced by a uniform parent; it is kept only in the legacy fig. 4/6 toy model for comparison. Valley splitting enters as valley-orbit (VO) coupling $\lambda(x) = |\lambda(x)| e^{i\phi(x)}$,

$$h_{VO}(x) = |\lambda(x)| [\cos \phi(x) \nu_x + \sin \phi(x) \nu_y] \otimes \sigma_0, \quad (3)$$

with splitting $E_v = 2|\lambda|$; a perfect interface has constant λ , and the phase ϕ is set by the interface position (Sec. 3).

2 The valley equivalence theorem

Theorem 1 (smooth-interface equivalence). *For any uniform VO phase ϕ and amplitude $|\lambda|$, the two-valley wire with inter-valley pairing (2), arbitrary valley-scalar terms (kinetic, μ , $V(x)$, Zeeman), and either valley-scalar Rashba SOC ($\alpha k \nu_0 \sigma_z$) or phase-locked Dresselhaus SOC ($\alpha k [\cos \phi \nu_x + \sin \phi \nu_y] \sigma_z$) is unitarily equivalent to two decoupled single-band wires with chemical potentials $\mu \mp |\lambda|$, pairing $\pm \Delta$, and the full SOC strength $\pm \alpha$ in each band.*

Proof. Let $U_\phi = e^{-i\phi\nu_z/2} \otimes \sigma_0$ act identically on every site. (i) Valley-scalar terms commute with U_ϕ . (ii) The VO term rotates to $|\lambda| \nu_x \sigma_0$ since $U_\phi (\cos \phi \nu_x + \sin \phi \nu_y) U_\phi^\dagger = \nu_x$. (iii) The pairing block transforms as $D \rightarrow U_\phi D U_\phi^T$ (not $U^\dagger$ on the right: D couples two creation operators). Because $\nu_z^T = \nu_z$ we have $U_\phi^T = U_\phi$, and using $\nu_x e^{-i\phi\nu_z/2} = e^{+i\phi\nu_z/2} \nu_x$,

$$U_\phi (\Delta \nu_x i\sigma_y) U_\phi^T = \Delta e^{-i\phi\nu_z/2} \nu_x e^{-i\phi\nu_z/2} i\sigma_y = \Delta \nu_x i\sigma_y. \quad (4)$$

The inter-valley pairing is exactly invariant under the valley rotation, for any ϕ . (iv) The phase-locked Dresselhaus SOC rotates, by the same algebra as (ii), to $\alpha k \nu_x \sigma_z$. After the rotation everything is diagonal in the ν_x eigenbasis $|\pm\rangle = (|+\rangle_v \pm |-\rangle_v)/\sqrt{2}$: the VO term becomes $\pm|\lambda|$ (a

chemical-potential shift $\mu \mp |\lambda|$, the pairing $\Delta \nu_x i \sigma_y \rightarrow \pm \Delta i \sigma_y$, and the Dresselhaus SOC $\rightarrow \pm \alpha k \sigma_z$. The sign of Δ is removed per band by $c \rightarrow e^{i\pi/2} c$, and the sign of α by $\sigma_z \rightarrow -\sigma_z$ (a spin rotation that leaves $E_Z \sigma_x$ invariant only combined with $\sigma_x \rightarrow \sigma_x$; explicitly, the unitary σ_x does this). Hence two decoupled standard wires at $\mu \mp |\lambda|$. $\square$

Numerical verification: `tests.py` (check 3) compares the full $8N$ spectrum against the decoupled-band reference for five phases $\phi \in \{0, 0.7, \pi/2, 2.6, 5.0\}$ and both SOC classes, enforcing agreement to $10^{-6} \mu\text{eV}$ on every eigenvalue (spot checks reach $\sim 3 \times 10^{-10} \mu\text{eV}$, i.e. solver precision). The practical consequence is the wedge picture of Fig. 4 of the main text: at $E_v \gtrsim 100 \mu\text{eV}$ one band can be placed in its topological window while the other is trivial, and a smooth interface is benign *regardless of the VO phase value*.

3 Atomic steps

3.1 The step phase $\Delta\varphi = 2k_0(a/4)$

The VO coupling is generated by the sharp interface potential $U_{\text{int}}(z)$, which scatters $+k_0 \rightarrow -k_0$ with momentum transfer $2k_0$:

$$\lambda \propto \int dz F^2(z) U_{\text{int}}(z - z_0) e^{2ik_0 z} \implies \arg \lambda \supset 2k_0 z_0, \quad (5)$$

where $F(z)$ is the smooth subband envelope and z_0 the interface height. A single-atomic step on a (001) surface changes z_0 by $a/4$ ($a = 5.431 \text{ \AA}$; the four inequivalent (001) atomic planes per cubic cell are spaced $a/4$). Hence

$$\Delta\varphi = 2k_0 \frac{a}{4} = 2 \cdot 0.85 \cdot \frac{2\pi}{a} \cdot \frac{a}{4} = 0.85\pi, \quad (6)$$

accidentally close to π because k_0 sits at 85% of the zone boundary in silicon. A double-height (bunched) step gives $2k_0(a/2) = 1.7\pi \equiv -0.3\pi \pmod{2\pi}$, which is *milder* — the basis of the step-bunching design rule in the main text. This is the standard interface-position dependence of valley-orbit coupling [Hosseinkhani–Burkard, PRB 100, 125309 (2019); Losert *et al.*, PRB 108, 125405 (2023)]; what is new here is its consequence under *inter-valley pairing*.

3.2 A step is a Josephson junction: band-projected pairing

Diagonalize the local VO term at fixed x : eigenvectors $|\pm\rangle_\phi = (|+\rangle_v \pm e^{i\phi} |-\rangle_v) / \sqrt{2}$ with energies $\pm|\lambda|$. Projecting the pairing (2) onto the local bands (pairing projects with the *conjugate* right basis, $D_{\text{band}} = W^\dagger D W^*$, $W = (|+\rangle_\phi, |-\rangle_\phi)$):

$$(W^\dagger \nu_x W^*)_{\eta\eta'} = \begin{pmatrix} e^{-i\phi} & 0 \\ 0 & -e^{-i\phi} \end{pmatrix}_{\eta\eta'} \implies D_{\text{band}} = \pm \Delta e^{-i\phi(x)} i\sigma_y. \quad (7)$$

Each band therefore sees a superconducting order parameter whose phase is minus the local VO phase. A VO phase step $\Delta\varphi$ at x_s is exactly a Josephson junction with phase drop $\Delta\varphi$ in the effective single-band topological wire. The gauge-invariant statement: the relative phase between $\lambda(x)$ and the (uniform) inter-valley pairing cannot be removed — a position-dependent rotation $U_{\phi(x)}$ leaves D invariant [Eq. (4) holds locally] and makes λ uniform, but generates a kinetic gauge field $k \rightarrow k - \frac{1}{2}(\partial_x \phi) \nu_z$, i.e. a δ -spike of ν_z -flux at the step; in the band basis ν_z is purely off-diagonal, so the spike both couples the bands and, within each band, reproduces the junction. The two pictures are related by the gauge choice; the spectrum is identical.

At $\Delta\varphi = \pi$ the order parameter changes sign: a topological wire with a π -junction hosts a Kitaev domain wall — an exact zero mode (numerically $E = 0.013\mu\text{eV}$, Table 1). Near $\Delta\varphi = 0.85\pi$ the bound state is deep. For a transparent short junction the Andreev bound state lies at $E_{ABS} = \Delta_t |\cos(\Delta\varphi/2)|$ with Δ_t the topological gap; Table 1 compares this zero-parameter formula with the lattice numerics.

Table 1: Single-step bound state vs the short-junction Andreev formula ($\Delta_t = 22.22\mu\text{eV}$, wedge point of Fig. 9; lattice values from `output/data/key_numbers.json`).

$\Delta\varphi/\pi$	0	0.10	0.25	0.40	0.50	0.65	0.85	0.95	1.00
lattice E_2 (μeV)	22.22	22.07	19.77	16.08	13.45	9.42	4.03	1.34	0.01
$\Delta_t \cos \frac{\Delta\varphi}{2} $	22.22	21.95	20.53	17.98	15.71	11.61	5.19	1.74	0.00

The formula tracks the numerics to $\lesssim 15\%$ for $\Delta\varphi \lesssim 0.5\pi$ and overestimates by 25–30% (relative to the lattice value) near 0.85π — the expected accuracy for an atomically abrupt junction (zero length but not in the Andreev limit $\Delta \ll$ all normal scales, and the bound state hybridizes with the wire-end MZMs at finite L). We therefore present the junction mechanism as an *analytically derived mapping* [Eq. (7), exact] with a *numerically quantified* bound-state depth, not as a closed-form spectrum.

3.3 Why net winding is irrelevant at realistic step densities

A same-sign staircase accumulates winding $\sum_s \Delta\varphi_s$; a sign-randomized control with identical $|\Delta\varphi|$ accumulates none. Fig. 9(a) of the main text shows the two ensembles are statistically indistinguishable: the damage is per-junction. Winding *does* matter in the smooth limit: a linear ramp $\phi(x) = qx$ is a uniform gauge field, pair-breaking only beyond $q \gtrsim 2|\lambda|/(\hbar v_F)$ -type thresholds (in-code generator, `fig9` key numbers: gap reduced by $\lesssim 25\%$ with no subgap state at $q \leq 4 \times 10^6 \text{ m}^{-1}$, destroyed by 1.6×10^7); at 50 nm step spacing the equivalent $q = 0.85\pi/50 \text{ nm} = 5.3 \times 10^7$ exceeds this, but the staircase damage at the same mean gradient is dominated by the discrete junction physics, as the control comparison proves.

4 Complete parameter tables

Baseline silicon parameters used throughout unless a figure row overrides them: $m^* = 0.19 m_e$, $g = 2.0$, $\Delta = 50 \mu\text{eV}$ (induced), $dx = 5 \text{ nm}$, $\alpha_{\text{demo}} = 0.05 \text{ eV \AA}$ (the “engineered” demonstration value), $\alpha_{\text{Si,opt}} = 10^{-3} \text{ eV \AA}$, $\alpha_{\text{InSb}} = 0.5 \text{ eV \AA}$. Disorder is onsite iid uniform in $[-W, W]$ with correlation length = dx (thresholds are lattice-convention dependent and quoted with dx). Bulk gaps minimize the lower BdG band over $n_k = 6001$ points unless noted. Seeds are `numpy default_rng` with the listed integer families; every random draw in the repository is seeded.

5 Convergence and sensitivity

All entries regenerate from `convergence.py` (`output/data/convergence.json`); Fig. 1 summarizes. The observables are the clean two-valley wedge gap and the single-step (0.85π) bound state at the Fig. 9 baseline ($\mu = 35 \mu\text{eV}$, $B = 1.5 \text{ T}$, $E_v = 150 \mu\text{eV}$), plus the hole-platform optimizer outputs.

The one headline number that moves under refinement is the measured-field Si:B center point: $9.9 \rightarrow 11.0 \mu\text{eV}$ bare ($9.1 \rightarrow 9.8\text{--}10.1$ renormalized) at finer optimizer grids — the production grid

Table 2: Electron-sector figures. $L = N dx$. “IV” = inter-valley pairing model [Eq. (2)]; “toy” = legacy intra-valley pairing.

Fig.	model	N (dx)	μ (μeV)	B (T)	α (eV Å)	valley	disorder	notes
1	1-band	400 (5 nm)	0	0.02–2 (40)	0.05	—	—	validation: fan, MZM, splitting
2	bulk	$n_E=6001$	–150..150 (201)	0.01–3 (181)	$10^{-3} / 0.5$	—	—	phase diagrams
3	bulk opt	$n_E=16, n_B=18$	0–145	≤ 2	$10^{-4}..0.5$	—	—	$\Delta(B)$ GL optional
4	toy	200 (5 nm)	–150..150 (31)	1.5	0.05	E_v : 0–300 (31), $\delta_{iv}=10$	—	seed [4-family]
5	1-band	L/dx (case)	0	1.5	case	—	W per case, 16 seeds	cases below
6	toy vs IV	300 (5 nm)	35	1.5	0.05	E_v : 0–300 (13)	—	8 seeds; steps $L_s=30, 100$ nm
7	2D strip	300×5 (5 $\times$ 10 nm)	subband windows	1.5	0.05	—	—	B along wire
9	IV	500 (5 nm)	35	1.5	0.05	$ \lambda =75$	steps: 6 spacings, 14 seeds	seed family [71,.]
12	RGF	$L=2 \mu\text{m}$	0	1.5	0.05	—	$W \in \{100, 200, 400, 800\}$	Ando leads, E -scan

Table 3: Fig. 5 disorder cases (per-case dx , L chosen for $k_{so}dx < 0.1$ and $L \gg \xi$ where affordable; 16 realizations each).

case	α (eV Å)	dx, L	W grid (μeV)
intrinsic Si	0.001	5 nm, 20 μm	0, 5, 10, 20, 50, 100, 200, 400
engineered Si	0.05	2.5 nm, 2 μm	0, 20, 50, 100, 200, 400, 800, 1600
strong SOC	0.15	2.5 nm, 6 μm	0, 20, 50, 100, 200, 400, 800, 1600

was $\sim 10\%$ *conservative*. The Pauli-scenario and Al numbers are grid-exact. Main-text ranges are quoted accordingly.

6 Parent superconductor models and the $g = 2$ optimum

6.1 What Δ is

Δ in the BdG Hamiltonian is a *static induced gap*: the infinite-coupling limit of the standard tunneling self-energy $\Sigma(\omega) = -\gamma(\omega + \Delta_p \tau_x) / \sqrt{\Delta_p^2 - \omega^2}$ evaluated at $\omega = 0$, with $\Delta_{ind} = \gamma \Delta_p / (\gamma + \Delta_p)$. In the production figures (i) field suppression enters through the parent, $\Delta_{ind} \leq \Delta_p(B)$, with $\Delta_p(B) = \Delta_0[1 - (B/B_c)^2]$ (GL, orbital-limited parents) or $\Delta_0 - \mu_B B$ (Pauli-limited caricature), and (ii) metallization enters through the quasiparticle weight $Z = 1 - \Delta_{ind}/\Delta_p$, which renormalizes $\alpha \rightarrow Z\alpha$ and $g \rightarrow Zg + (1 - Z)g_p$ ($g_p = 2$). Both caricatures are stated wherever used. The full dynamical self-energy is solved as a control in Sec. 9.1: it lowers the optimal gaps 24–30% below the static-renormalized values (which therefore act as upper bounds) and agrees with the static model at matched effective coupling to $\lesssim 7\%$ — the caricature’s direction and approximate magnitude are confirmed.

6.2 Connection to Cole–Das Sarma–Stanescu

Ref. [Cole *et al.*, PRB 92, 174511 (2015)] showed that strong semiconductor–superconductor coupling buys induced-gap size at the price of semiconductor properties (g, α), so an intermediate coupling optimizes the topological gap. Our $g = 2$ ceiling is the sharp small- g limit of that statement: with $E_Z(2\text{ T}) = 116 \mu\text{eV}$ fixed by $g = 2$, the topological condition $E_Z > \sqrt{\Delta_{ind}^2 + \mu^2}$ cannot be met at all for $\Delta_{ind} \geq 116 \mu\text{eV}$, and maximizing the gap over $(B, \Delta_{ind} \leq \Delta_p(B), \mu)$ under GL suppression yields an interior optimum $\Delta_0 \approx 98 \mu\text{eV}$ (main-text Sec. 3.2) — i.e. for silicon electrons the Cole optimum is not merely beneficial but *mandatory*, and it caps the achievable gap at a value disorder then destroys. The hole platform escapes through g_z up to 4.2, which moves the ceiling to $\sim 240 \mu\text{eV}$ at 2 T.

Table 4: Hole-sector figures. Parent models in Sec. 6. Optimizer: gap maximized over $B \leq B_{\max}(n_B=7)$, $\Delta_{ind} \in [8, \Delta_p(B)]$ ($n_\Delta=7$), $\mu \in [0, 160] \mu\text{eV}$ ($n_\mu=14$), $n_k=2501$ (convergence: Sec. 5).

Fig.	model	parameters
8	bulk opt	(α, g) box: $\alpha \in [0.01, 0.3] \text{ eV \AA}$ (log, 20), $g \in [1.2, 4.0]$ (20); $m^* = 0.25$; center point $(\alpha, g) = (0.06, 2.2)$; parents Si:B-Pauli / Si:B-measured / Al; metallization $Z = 1 - \Delta_{ind}/\Delta_p$, $\alpha \rightarrow Z\alpha$, $g \rightarrow Zg + (1-Z)g_{\text{parent}}$, $g_{\text{parent}} = 2$. Finite-wire check at center: $N=1200$, $dx=2.5 \text{ nm}$, 10 seeds, $W \leq 800 \mu\text{eV}$.
10	4-band LK	fin $W_y \times W_z = 10 \times 12 \text{ nm}$ hard wall, grid $n_y \times n_z = 11 \times 13$; $\gamma_1, \gamma_2, \gamma_3 = 4.285, 0.339, 1.446$, $\kappa = -0.42$ (Lawaetz); $E_z \in \{2, 5, 10, 15, 20, 30, 40, 50\} \text{ MV/m}$; bare + orbital Zeeman; no strain, no cubic J^3 terms (stated caricature).
11	orientation	$\theta \in [0^\circ, 90^\circ]$ (8), $\phi_B \in [0^\circ, 180^\circ]$ (13); LK tensor at $E_z = 10 \text{ MV/m}$ ($\alpha=0.054$, $ g =(0.5-1.1, g_y, 2.4-4.2)$, SOC axis $\hat{y}$) vs empirical tensor ($\alpha=0.06$, $g=(2.1, 2.35, 2.7)$, SOC axis $0.41\hat{y} + 0.91\hat{z}$); parents: thick Si:B (isotropic GL, $B_{c2}=0.4 \text{ T}$) and Al film ($B_{c\parallel}=2 \text{ T}$, $B_{c\perp}=0.1 \text{ T}$); tilted-gap $n_k=500$ with the Osca-Ruiz-Serra criterion.
13	convergence	Sec. 5; engine <code>convergence.py</code> .

Table 5: Discretization and length convergence (μeV).

	$dx \text{ (nm)}, L=2.5 \mu\text{m}$				$L \text{ (}\mu\text{m)}, dx=5 \text{ nm}$		
	10	5	2.5		1.25	2.5	5.0
clean gap	22.165	22.221	22.234	clean	21.566	22.221	22.239
step 0.85π	4.037	4.034	4.033	step	4.149	4.034	4.034
π -step E_0	0.013	0.013	0.013	ens. median ^a	2.537	0.648	0.083

^a50-nm same-sign Poisson staircase, 14 seeds. The L -dependence is physical (Anderson localization; transport section of the main text): ensemble medians are quoted only as L -tagged bounds.

7 Quasiparticle-poisoning estimate

The equilibrium quasiparticle fraction of the parent at temperature T is $x_{qp} \simeq \sqrt{2\pi k_B T / \Delta_p} e^{-\Delta_p / k_B T}$. Requiring $x_{qp} \leq 10^{-6}$ with the Si:B parent gap *at the operating field* ($\Delta_p(B^*) = 29\text{--}33 \mu\text{eV}$ for the measured-field design) gives $T_{\text{eff}} \leq 25\text{--}29 \text{ mK}$; the same criterion for the Al reference point of `qp_poisoning.py` ($\Delta_p = 150 \mu\text{eV}$ at $B = 1.0 \text{ T}$, GL) gives 130 mK . Since realistic electron temperatures in superconducting devices rarely reach 25 mK , the measured-field all-silicon stack is classified as a detection platform (ZBP spectroscopy, gap hardness, the miscut test) rather than a parity-coherent qubit, absent parent-gap engineering. All numbers in `qp_poisoning.py`.

8 Luttinger–Kohn extraction

The fin cross-section (y, z) is solved in the 4-band LK Hamiltonian (hard-wall, grid 11×13 , $\gamma_1 = 4.285$, $\gamma_2 = 0.339$, $\gamma_3 = 1.446$, $\kappa = -0.42$) with vertical field E_z ; the lowest Kramers doublet is followed in k_x to extract m^* (curvature), α (linear-in- k splitting), the SOC axis (spin direction of the split states), and the g -tensor (bare $2\kappa\mu_B \mathbf{B} \cdot \mathbf{J}$ plus orbital terms, evaluated as the field-derivative of the doublet splitting along the three axes). Validation: at $E_z = 0$ the model reproduces the exact bulk [100] heavy/light masses, and the predicted spin-orbit lengths $l_{so} = \hbar^2 / (m^* \alpha) = 32\text{--}101 \text{ nm}$ overlap the measured FinFET window ($\sim 20\text{--}60 \text{ nm}$). Limits: no strain, no cubic (J^3) Zeeman, no device electrostatics — the known shortfall is $g_x \in [0.4, 1.1]$ across a seven-geometry bracket ($7\text{--}16 \text{ nm}$ sections, $3\text{--}30 \text{ MV/m}$) versus measured $g_{xx} \approx 1.9\text{--}2.3$; all platform numbers are therefore carried under both tensors, and the 6-band+Poisson resolution is the top open theory item.

Table 6: Statistical and parametric sensitivity.

seeds n	7	14	28	56
running median (μeV)	0.724	0.648	0.583	0.638
56-seed bootstrap 95% CI: $[0.516, 0.810] \mu\text{eV}$				
μ (μeV)	15	25	35	45/60
clean / step	9.7 / 1.9	16.7 / 3.0	22.2 / 4.0	24.6–27.6 / 5.0–6.2
E_v (μeV)	100	150	225	300
clean / step	11.6 / 2.6	22.2 / 4.0	exits topological window at fixed μ^b	

^bAt fixed $\mu = 35 \mu\text{eV}$, $B = 1.5 \text{ T}$ the lower band leaves its topological window for $E_v > 212 \mu\text{eV}$ (analytic; grid points $\geq 225 \mu\text{eV}$ are outside); single-valley topology at larger E_v requires re-centering μ (main-text Fig. 4 wedge).

The quoted values at $E_v \geq 225$ are trivial gaps and are excluded from any protection claim.

Table 7: Hole-platform optimizer convergence and parent-model sensitivity (gap in μeV ; (r) = metallization-renormalized).

grid $n_B \times n_\Delta \times n_\mu$	$5 \times 5 \times 10$	$7 \times 7 \times 14$	$10 \times 10 \times 20$	$14 \times 14 \times 28$
Si:B measured	8.83	9.92	11.01	11.01
Si:B measured (r)	7.93	9.14	10.13	9.82
Si:B Pauli	30.62	30.62	30.62	30.62
Si:B Pauli (r)	19.68	19.68	19.68	20.21

k -grid n_k : $\leq 0.022\%$ drift from 501 to 12001 at all three operating points.

Parent model: $\Delta_0 \times \{0.8, 1, 1.2\}$, $B_{c2} \in \{0.3, 0.4\} \text{ T} \Rightarrow$ measured-field center 8.0–10.6 (7.6–10.0 r).

9 Parent and disorder realism controls

Four controls (review round 5) test the headline numbers against the caricatures they rest on. All re-generate from `realism.py`, `orbital.py`, `pairing_mix.py`, `morphology.py` (ledgers `realism/orbital/pairing_morphology` in `key_numbers.json`).

9.1 Dynamic tunneling self-energy

The static Δ is replaced by the full frequency-dependent BCS self-energy $\Sigma(\omega) = -\Gamma(\omega + \Delta_p \tau_x) / \sqrt{\Delta_p^2 - \omega^2}$; quasiparticle energies solve $\det[H(k) + \Sigma(\omega) - \omega] = 0$ (batched eigenvalue crossing scan; α , g , m^* renormalization is automatic through the ω -linear term). Validation: static limit ($\Delta_p \rightarrow \infty$, $\Sigma \rightarrow -\Gamma\tau_x$) matches `bulk_gap_ueV` to 0.07%; (n_k, n_ω) doubling changes results by 0.001%. The topological criterion is exact at $\omega = 0$: $E_Z^2 > \Gamma^2 + \mu^2$.

Table 8: Dynamic topological gap (μeV) vs tunnel coupling Γ at the three parent operating points ($\mu = 0$, empirical-center hole tensor).

Γ (μeV)	3	6	10	15	20	30	45	70	100	best (Γ^*)	static@ $\Delta_{ind}(\Gamma^*)$	static bare
Si:B meas. (0.33 T)	2.7	5.0	7.6	3.9	0.6	0	0	0	0	7.6 (10)	7.4	11.0
Si:B Pauli (1 T)	2.7	5.0	7.6	10.4	12.7	14.9	7.3	0	0	14.9 (30)	15.2	30.6
Al (1.7 T)	2.7	5.0	7.9	11.1	13.9	18.7	24.2	15.3	2.9	24.2 (45)	22.9	50.4

The dynamic optimum is $1.4\text{--}2.1\times$ below the static *bare* numbers, sits 24–30% below the static- Z *renormalized* caricature (10.1 / 19.7 / 34.4), and agrees with the static model evaluated at the

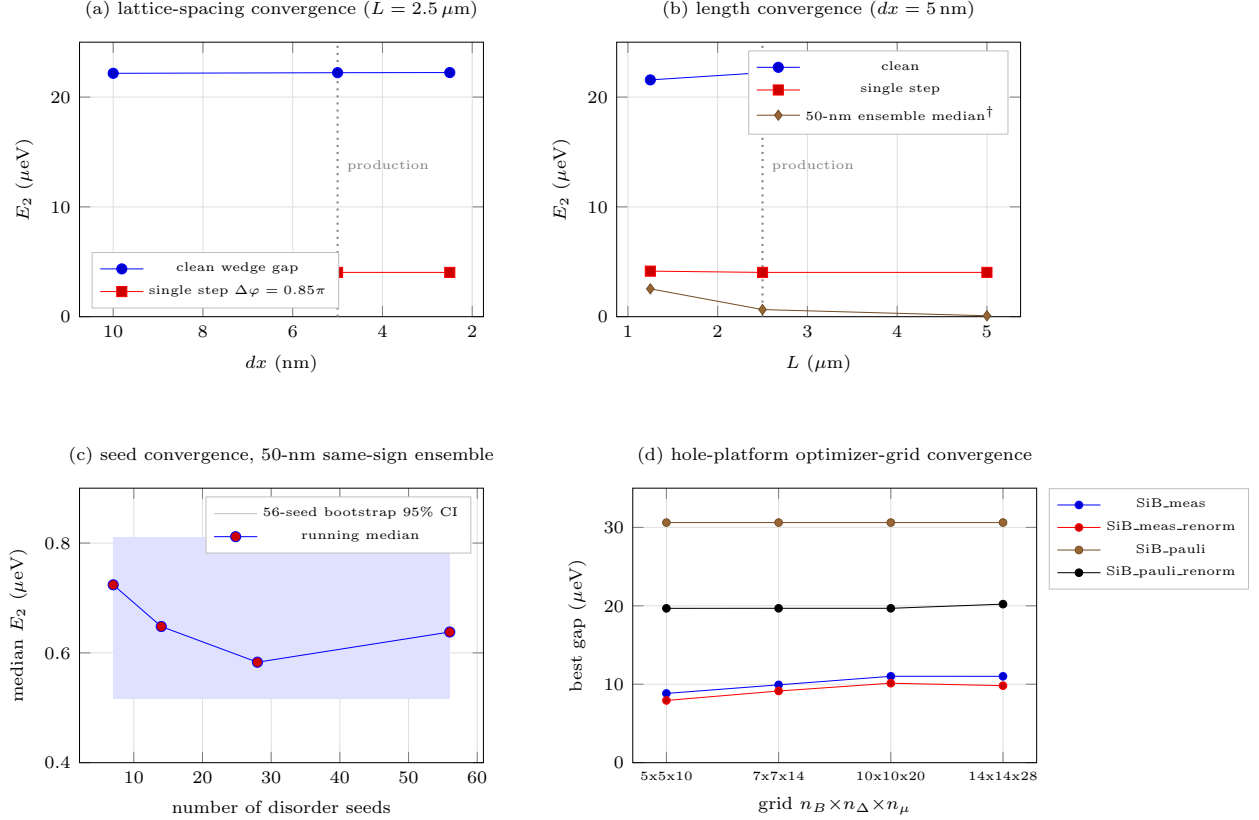


Figure 1: Convergence summary (data: `convergence.json`; rendered by `tools/fig13-pgf.py`, equivalently `convergence.py` with `matplotlib`). [†]The L -dependence of the step-ensemble median in panel (b) is physical (Anderson localization), not a discretization artifact; see Table 5, note *a*.

matched effective induced gap $\Gamma^*\Delta_p/(\Gamma^* + \Delta_p)$ to $\lesssim 7\%$ — the static- Z treatment is directionally validated, and all main-text gap ranges now carry the dynamic correction as the better central estimate. Dynes broadening adds a temperature floor rather than a gap penalty: subgap DOS fraction γ_D/Δ_p dominates the thermal quasiparticle budget below 37–71 mK (Si:B, $\gamma_D/\Delta_p = 10^{-4}$ – 10^{-2}) and 71–136 mK (Al), so 50 mK operation requires $\gamma_D/\Delta_p \lesssim 10^{-3}$ for Si:B parents. Correlated parent-gap inhomogeneity ($\Delta(x)$, 50 nm correlation) is benign at the 10–25% level (median gap 88–97% of clean) and halves the gap at 50%.

9.2 Orbital magnetic coupling

Peierls phases of the perpendicular field component on a 2D strip (validated against `build_wire_2d` to 3×10^{-13} μeV at $B_\perp = 0$; exact Hermiticity and PHS). At the preferred tilted Si:B operating point ($B = 0.33$ T, $\theta \approx 39^\circ$, $\Delta_{ind} = 11.3$ μeV) the orbital gap suppression is **0.02–0.04%** for fin widths 7–16 nm and 0.3% at 40 nm (scaling $\propto W^{1.5}B_\perp^2$; the 10% level extrapolates to $W \sim$ several hundred nm). Even a full 1 T out-of-plane field costs only 0.3 / 1.4 / 8.9% at $W = 10/20/40$ nm, with the zero mode and end localization intact. With μ at the lowest subband edge the transverse subband spacing (0.7–6 meV) dwarfs the orbital matrix elements; omitting orbital effects does not alter any orientation-map conclusion at the μeV scale.

9.3 Pairing-channel mixing

$D(\eta) = \Delta[\sqrt{1 - \eta^2}(\nu_x \otimes i\sigma_y) + \eta(\nu_0 \otimes i\sigma_y)]$ interpolates from the physical inter-valley channel to pure intra-valley pairing at preserved total weight ($\eta = 0$ reproduces the production builder to $2 \times 10^{-13} \mu\text{eV}$). The dilution intuition is *wrong* in the direction favorable to our claims: the valley rotation that removes a VO phase step maps $\nu_0 \otimes i\sigma_y \rightarrow e^{-i\phi\nu_z} \otimes i\sigma_y$, so the intra channel inherits the step as a pairing-phase junction — the single-step bound state deepens monotonically ($4.03 \rightarrow 3.12 \rightarrow 1.68 \mu\text{eV}$ at $\eta = 0, 0.1, 0.3$; $2\times$ by $\eta^* \approx 0.24$; near-zero junction modes at $\eta = 1$). Staircase *ensembles* conversely heal with admixture (median E_2 : $0.57/1.8/4.5 \mu\text{eV}$ at $\eta = 0/0.1/0.3$; the phase-blind intra channel supplies a pairing floor), so the $\eta = 0$ ensemble gaps quoted in the main text are conservative. Physically $\eta \ll \eta^*$: an intra-valley pair carries momentum $2k_0$, which a momentum-conserving uniform parent cannot supply; intra-valley pairing requires interface-roughness Fourier weight at $2k_0$ — the same small parameter that generates the VO coupling itself.

9.4 Step morphology and correlated electrostatic disorder

Morphology (gamma-distributed terraces, shape $k = 4$; miscut θ_m sets the mean terrace $s = (a/4)/\tan\theta_m$; wire at angle χ to step edges; generator validated against the fig. 9 pipeline to $5 \times 10^{-7} \mu\text{eV}$ per seed). Reported as $p_5/p_{50}/p_{95}$ of E_2 (μeV) and the failure fraction $f_{<1} = P(E_2 < 1 \mu\text{eV})$: miscut scan ($\chi = 90^\circ$): 0.05° : $0.5/2.0/3.1$, $f_{<1} = 0.25$; 0.1° : $0.3/1.1/2.8$, 0.33 ; 0.2° : $\sim 0/0.005/0.02$, 1.00 ; 0.5° : $0/0/0.001$, 1.00 . Orientation ($\theta_m = 0.1^\circ$): $\chi = 90/45/20/10/5^\circ$ medians $1.5/1.7/2.2/3.2/3.8$ with $f_{<1}$ falling $0.33 \rightarrow 0$. Finite ramp widths $\leq 10 \text{ nm}$ and VO-amplitude dips (to 30% over 3 nm) shift medians by only 10–15% (the latter strictly worsening); double-step bunches ($\Delta\varphi \equiv -0.3\pi$) give $2.7/5.9/10.2$ with $f_{<1} = 0$ at the same height budget. Rare-event tails, not medians, are binding: at realistic $0.05\text{--}0.1^\circ$ miscuts 25–33% of realizations fail even where the median looks viable — yield is the constraint. Design rule: wires along step edges ($\chi \lesssim 10^\circ$) on $\leq 0.05^\circ$ substrates, or deliberately step-bunched templates.

Correlated disorder (Gaussian random field $\mu(x)$, correlation length λ_c , vs iid at matched RMS; hole operating point, clean $E_2 = 31.3 \mu\text{eV}$): at $50 \mu\text{eV}$ RMS the median collapses to $6.8/4.8/3.5 \mu\text{eV}$ for $\lambda_c = 25/50/100 \text{ nm}$ while the iid baseline retains $28.9 \mu\text{eV}$ — *the iid model overstates robustness by 4–8 $\times$ in gap at fixed RMS*, making gate-potential smoothness at the few-tens-of- μeV level the binding fabrication requirement. α fluctuations (20%, $\lambda_c = 50 \text{ nm}$) are negligible (median 29.6); g fluctuations (20%) cost $\sim 12\%$ (median 27.5). (Caveat: at RMS = $100 \mu\text{eV} > E_Z = 64 \mu\text{eV}$ the wire fragments and E_2 no longer measures a topological gap; 10 seeds per cell.)

10 Reproducibility

Every figure and quoted number regenerates from the committed code: `run_analysis.py` (figures 1–11), `transport.py` / `transport_valley.py` (RGF, invariant, valley transport), `lk_holes.py`, `qp_poisoning.py`, `convergence.py` (Sec. 5), `realism.py`, `orbital.py`, `pairing_mix.py`, `morphology.py` (Sec. 9), `tests.py` (assertion suite covering the equivalence theorem, gauge covariance, analytic dispersions, RGF agreement with closed-chain spectra to 10^{-13} , and regression tests for every bug found in review). The numerical ledger is `output/data/key_numbers.json`; scan checkpoints allow resuming every long run. The repository is public: <https://github.com/wtrevena/feasibility-of-creating-majorana-modes-in-silicon-nanowires>. Archival packaging (commit hash, environment, per-figure commands) is described in `README.md`; randomness is fully seeded

(seed families are listed in Tables 2–4).